COLLEGE OF ENGINEERING, PUNE

(An Autonomous Institute of Government of Maharashtra.)

END Semester Examination

(CT-20014) Data Structures and Algorithms – II

Course: B.Tech , Semester IV Branch: Computer Engineering

Academic Year: 2023-2024

Max.Marks:20

Duration: 1 Hours

Date: 17/02/2024

Instructions: **Student MIS No.**

1. Figures to the right indicate the full marks.
2. Mobile phones and programmable calculators are strictly prohibited.
3. Writing anything on question paper is not allowed.
4. Exchange/Sharing of stationery, calculator etc. not allowed.
5. Write your MIS Number on Question Paper

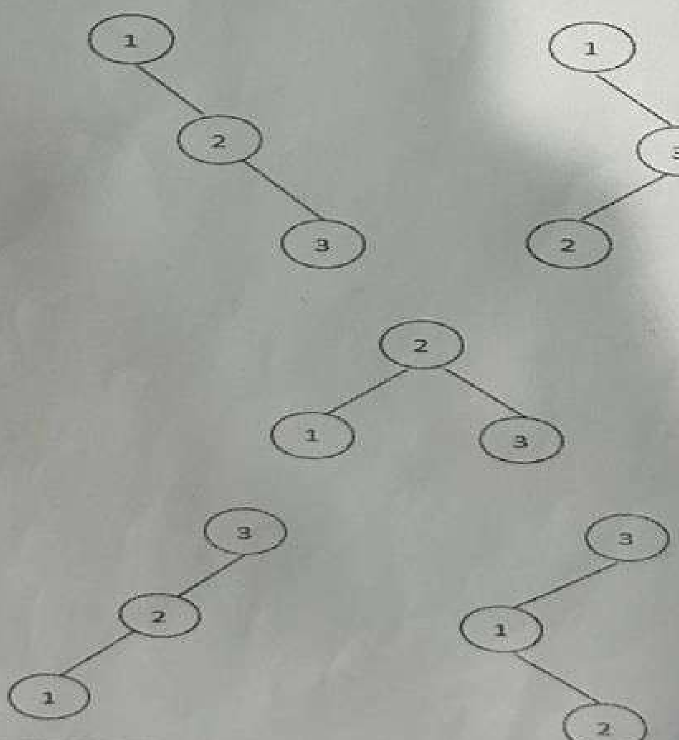
			Marks
Q 1	**a**	Given an integer n, explain how many numbers of structurally unique binary search trees will be there which has exactly n nodes of unique values from 1 to n.	**2**

Ans:

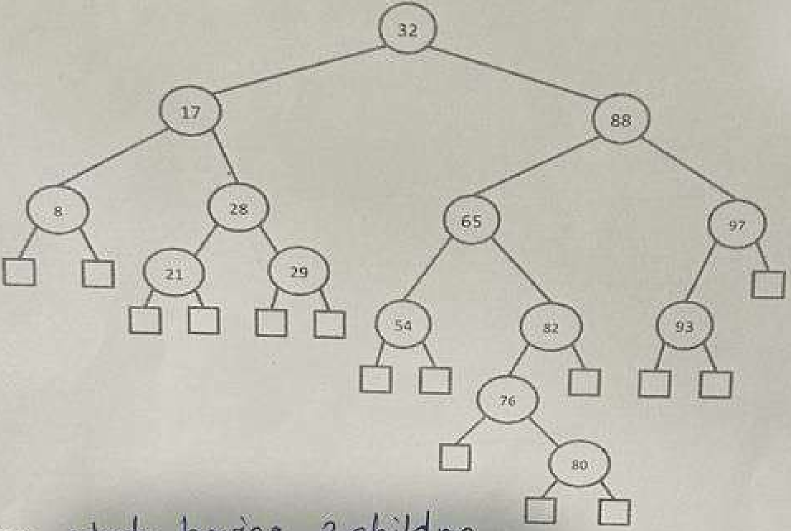
1 → **(2n)! / ((n+1)!*n!)** or $\frac{{}^{2n}C_n}{n+1}$

Eg n=3

1 → **T=5**



	b	If number of nodes 'n' are given in binary tree then minimum and maximum height of the tree will be ________ and ________ respectively.	2
1	→	**Ans: min log(n+1)-1**	
1	→	**Max: n-1**	
Q 2	a	Consider the binary search tree below. Draw the binary search tree obtained after performing remove (88).	2

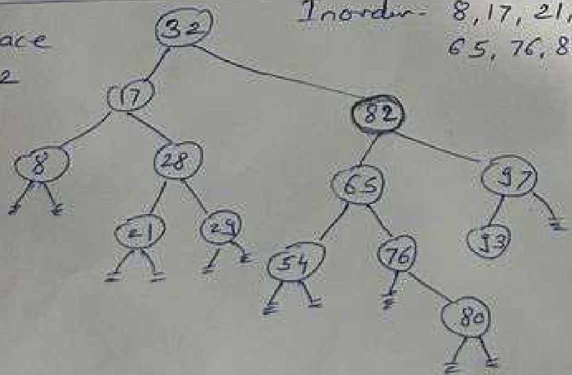


Ans: Node having 2 children

1. Largest from LST replace or Inorder succ predecessor.
2. Smallest from RST replace or Inorder successor.

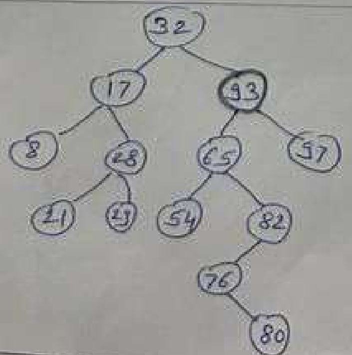
Inorder - 8, 17, 21, 28, 29, 32, 54, 65, 76, 80, 82, 88, 93, 97

1. Replace with 82



→ 1

2. Replace with 93



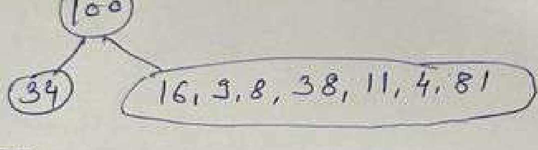
→ 1

b	Let the preorder traversal sequence of binary tree T be 100, 34, 16, 9, 8, 38, 11, 4, 81 and postorder traversal sequence be 34, 9, 11, 4, 38, 81, 8, 16, 100. If all the non-leaf nodes of T have two children, identify T.	3

① In preorder 1st element is root

∴ 100 Root

② In preorder next to 100 is 34



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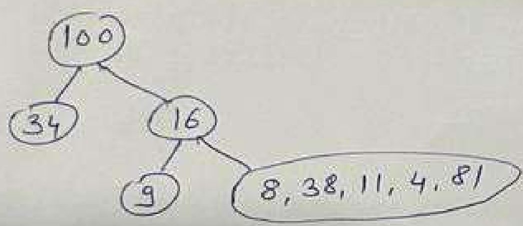

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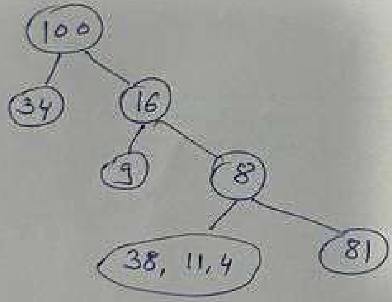
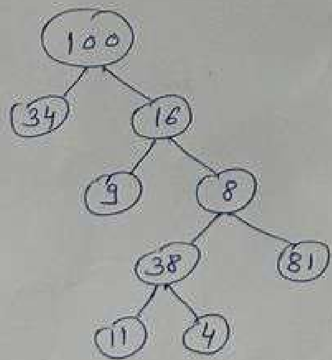


⑤



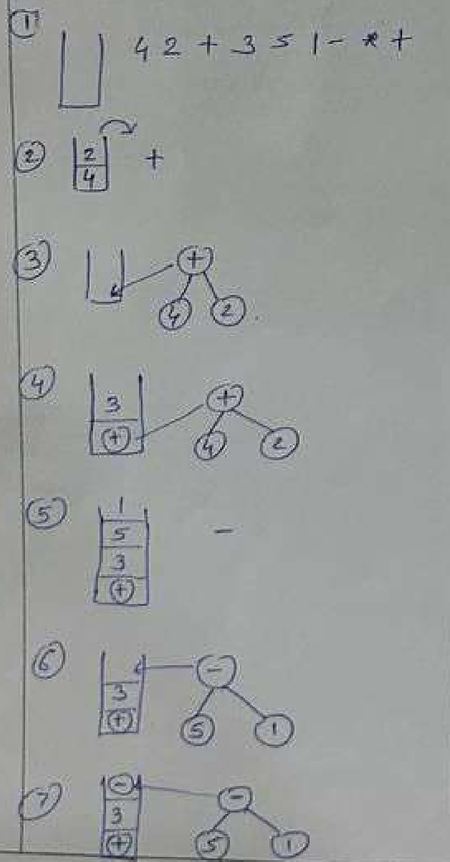
Q 3 **a** Write the output of the following function: **2**

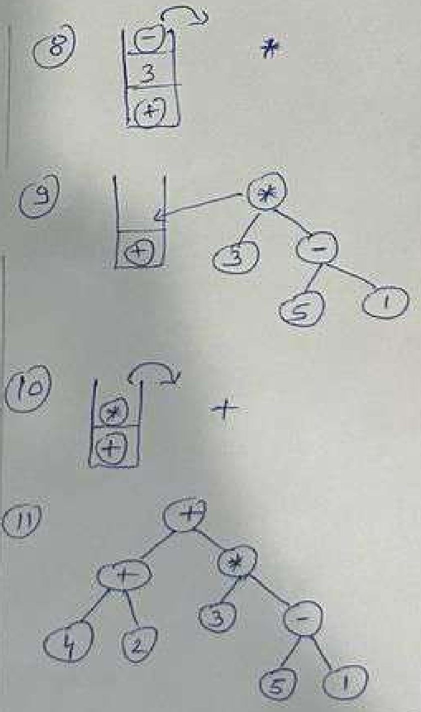
```
void function1(tree t)
{
    node *p;
    queue q;
    qinit(&q);
    enq(&q, t);
    while(!qempty(&q))
    {
        p=deq(&q);
        printf("%d",p->data);
        if(p->left)
        enq(&q, p->left);
        if(p->right)
        enq(&q, p->right);
    }
    printf("\n");
}
```

The tree t is a binary search tree, having data inserted in the given order: 75, 68, 89, 21, 72, 80, 96. Make suitable assumptions wherever required.

Ans: Levelwise Print
75, 68, 89, 21, 72, 80, 96.

b Explain construction of Expression tree to evaluate given postfix expression - **3**
Input: s = ["4","2","+","3","5","1","-","*","+"]





Ans - [(4+2)]+[3*(5-1)]
= 18

Q4

a Write a function to display the path from root to the farthest leaf of a binary tree of integers. Assume linked implementation of the tree. Also assume all required data structures are available. **3**

Ans:

```
void longestPath(BST t){
if(t == NULL)
return;
printf("%d ",t->data);
if(height(t->left) > height(t->right))
longestPath(t->left);
else
longestPath(t->right);
return;
}
```

b Write an iterative function in C to insert node in a BST using linked list. Given a root node reference of a BST and a key, insert the node with the given key in the BST. **3**

```
void insert (tree* t, int data)
{
    node *p =*t, *q;
        while (p!=NULL)
        {
            q=p;
            if(p->data==data)
            {
                return;
            }
            else if(data < p->data)
            p=p->left;
            else if(data > p->data)
            p=p->right;
        }
    if(data < q->data)
        q->left = create_node(data);
    else q->right=create_node(data);
}
```